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Process Design Kit and Design Automation for Flexible Hybrid Electronics

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***Abstract*—High-performance low-cost flexible hybrid electronics (FHE) are desirable for applications such as internet of things and wearable electronics. Carbon nanotube (CNT) thin-film transistor (TFT) is a promising candidate for high-performance FHE because of its high carrier mobility, superior mechanical flexibility, and material compatibility with low-cost printing and solution processes. Flexible sensors and peripheral CNT-TFT circuits, such as decoders, drivers, and sense amplifiers, can be printed and hybrid-integrated with thinned (<50 μm) silicon chips on soft, thin, and flexible substrates for a wide range of applications, from flexible displays to wearable medical devices. Here, we report (1) a process design kit (PDK) to enable FHE design automation for large-scale FHE circuits and (2) solution process-proven intellectual property blocks for TFT circuits design, including Pseudo-Complementary Metal-Oxide-Semiconductor (Pseudo-CMOS) flexible digital logic and analog amplifiers. The FHE-PDK is fully compatible with popular silicon design tools for design and simulation of hybrid-integrated flexible circuits.**

***Keywords*—Flexible hybrid electronics, thin-film transistors, thinned silicon chips, printed electronics**

Introduction

Flexible and printable electronics have exponential growth in performance and cost reduction in recent years, which enables new applications such as disposable sensors, radio frequency identification (RFID) tags, and low-cost internet of things (IoT) [2, 3]. To design a large-scale (>1,000 transistor count) digital and analog flexible thin-film transistor (TFT) circuitry that is manufacturable with low-cost printing and solution processes, however, is another story. Because of large printing process variations, as well as the lack of complementary and reliable TFTs, complex TFT circuits often suffer from high static power consumption and low circuit yields, particularly in low-supply voltages, which make TFT circuits design a significant challenge.

On the other hand, the hybrid integration of thinned silicon chips and printable TFTs on the same flexible substrate emerges as a viable solution for a wide range of applications. The flexible printed sensors and TFT circuits that benefit from low manufacturing costs and conformal form factors can be an ideal complement to thinned silicon chips for designing a flexible IoT or wearable device. We have previously reported our progress of developing flexible hybrid electronics (FHE)-process design kit (PDK) [4, 5] that includes (1) simulation models of printable passive and active devices, such as TFTs, resistors, inductors, and capacitors; (2) design rules of printable devices validated using specific printing or solution-process technologies; (3) multiphysics simulation models that can capture the interaction among electrical-thermal-mechanical performance; and (4) co-design capability for both printed electronics and thinned silicon chips, using the same design environment and electronic design automation (EDA) toolsets to ease the FHE design process.

In this article, we report our latest progress regarding FHE-PDK to include (1) models of carbon nanotube (CNT), organic and indium-gallium-zinc-oxide (IGZO) TFTs, CNT resistors, and parallel-plate capacitors; (2) experimentally validated design rules and physical design verification capability, including design rule checking (DRC), layout versus schematics checking, and layout parasitics extraction (LPE); and (3) low-voltage (3V) TFT digital and analog design intellectual property (IP) shown in Fig. 1. The technology files, simulation program with integrated circuit emphasis (SPICE) simulation models, and parameterized cells (P cell) are fully compatible with mainstream integrated circuit (IC)-centric EDA toolsets for silicon CMOS design, which makes FHE-PDK an ideal choice for codesign, cosimulate, and coverify FHE design containing both printed TFTs and silicon chips. The current PDK also supports the printed circuit board (PCB)-centric tool such as Mentor Graphics *HyperLynx* for DRC, signal and power integrity analysis.

Device Modeling

A. Compact Thin-Film Transistor Models

Several TFT compact models targeting specific technologies [6] have been developed in the past, which may not be generalized for other technologies. Some studies focus only

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on modeling the DC behavior [7], which does not support transient simulations. In this article, we present a unified compact model of TFTs covering DC, AC, and technology scaling factor for supporting FHE design, and also validate the proposed models against measurement results using three TFT technologies: CNT-TFTs, IGZO TFTs, and organic TFTs [4]. The model-measurement validation using current-voltage (I-V) curves is shown in Fig. 2. Based on this model, we further perform circuit-level validation based on fabricated *Pseudo-CMOS* [1] inverters, sequence generators, and ring oscillators. The proposed model is implemented using Verilog-A to perform SPICE cosimulations with thinned silicon CMOS circuitry, thus enabling FHE designers to explore a variety of design and integration options for TFT-CMOS flexible hybrid circuits and evaluate the performance for a wide range of commercial and military applications from sensing, display, IoT, and health care to wearable medical devices [4].

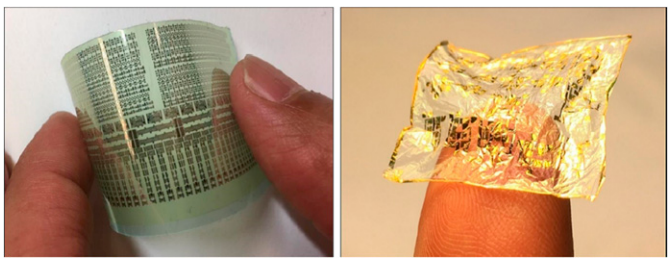

Fig. 1. Circuit design IP blocks, including *Pseudo-CMOS* [1] digital logic, analog amplifies, and passive elements (resistor/capacitor). (Left) solution-processed CNT-TFT circuits on the 10-μm flexible substrate for bending test and (right) CNT-TFT circuits on the 1-μm substrate.

B. Resistor Models

In addition to TFTs, the CNT film can also be used to realize a linear resistor, and the manufacturing process of a CNT resistor is fully compatible with CNT-TFTs. The CNT resistors have a wide range of applications, such as linear and transimpedance amplifiers and low-pass filters. To further study the CNT resistor linearity and process variations, we fabricated and characterized 864 CNT resistors on a flexible substrate shown in Fig. 3. With 40 μm resistor width W, the length L is varied from 2 to 100 μm and the measured I-V curves are shown in Fig. 4. CNT resistors show a good linearity when the length is greater than 10 μm, when compared against a linear fitting curve. Fig. 5 further reveals the process variations of the CNT resistors, where a total of 96 test samples are included for each test case of the resistor lengths L = 10 μm, 50 μm, and 100 μm, and a total of 384 test samples are included for the resistor length L = 20 μm. Although the histograms follow Gaussian or normal distributions, we can learn that the standard deviations of the CNT resistance reduce when the resistor length L increases. The sheet resistance of the CNT film is around 137 kΩ for L = 10 μm and decreases to 103 kΩ for L = 100 μm. For the CNT resistors, the resistance values can be readily calculated using $R_{CNT} = R_{CH} \times L/W_R + R_C/W_R$, where R_{CNT} is the total resistance in KΩ, R_{CH} is the channel resistance in KΩ, R_C in KΩ is the contact

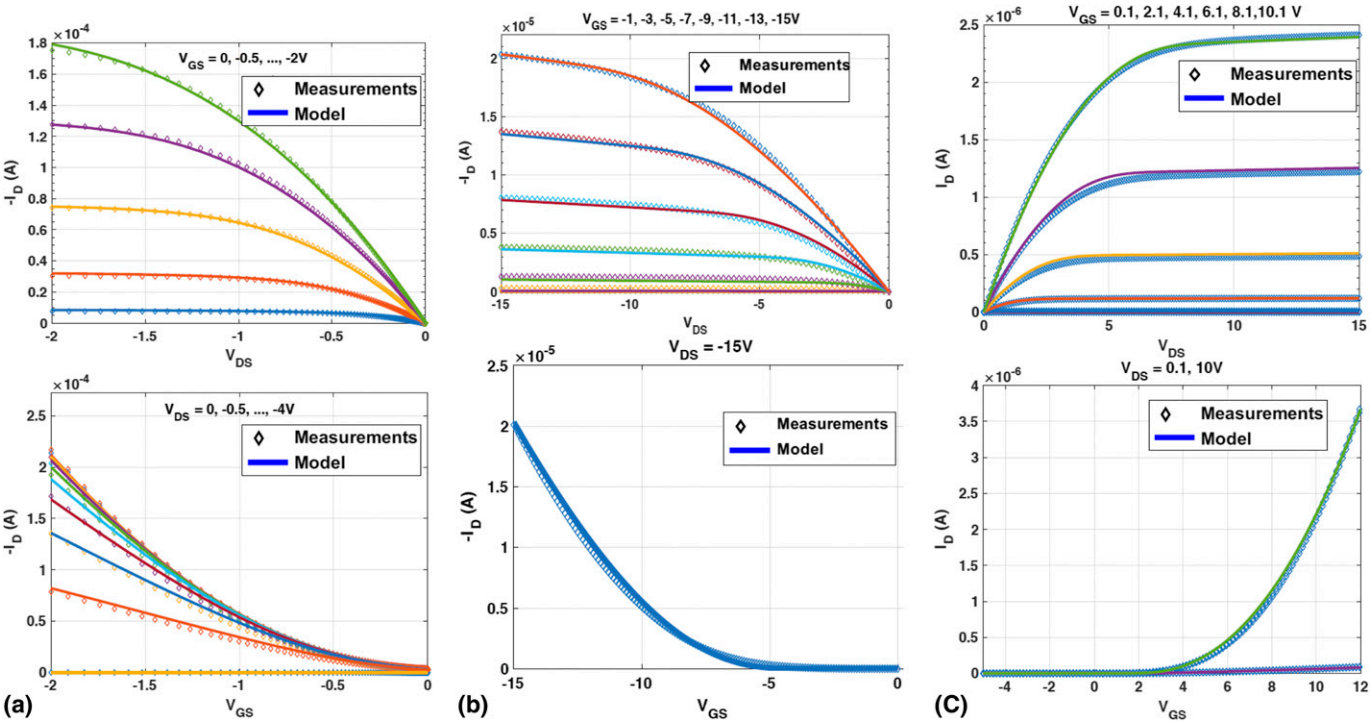

Fig. 2. Measured (dotted) and fitted (solid) current-voltage (I-V) curves for (a) CNT-TFTs (W = 125 μm and L = 25 μm), (b) organic TFT (W = 5,000 μm, L = 10 μm), and (c) IGZO TFT (W = 30 μm, L = 20 μm) [4].

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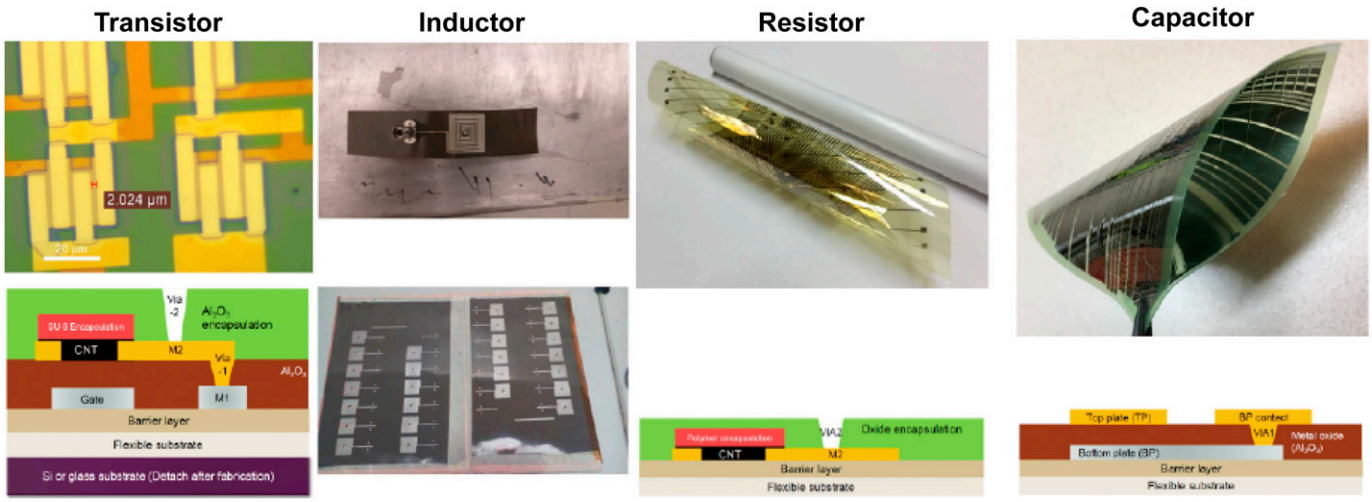

Fig. 3. Flexible active and passive devices including CNT-TFTs, spiral inductors, CNT resistors, and parallel-plate capacitors.

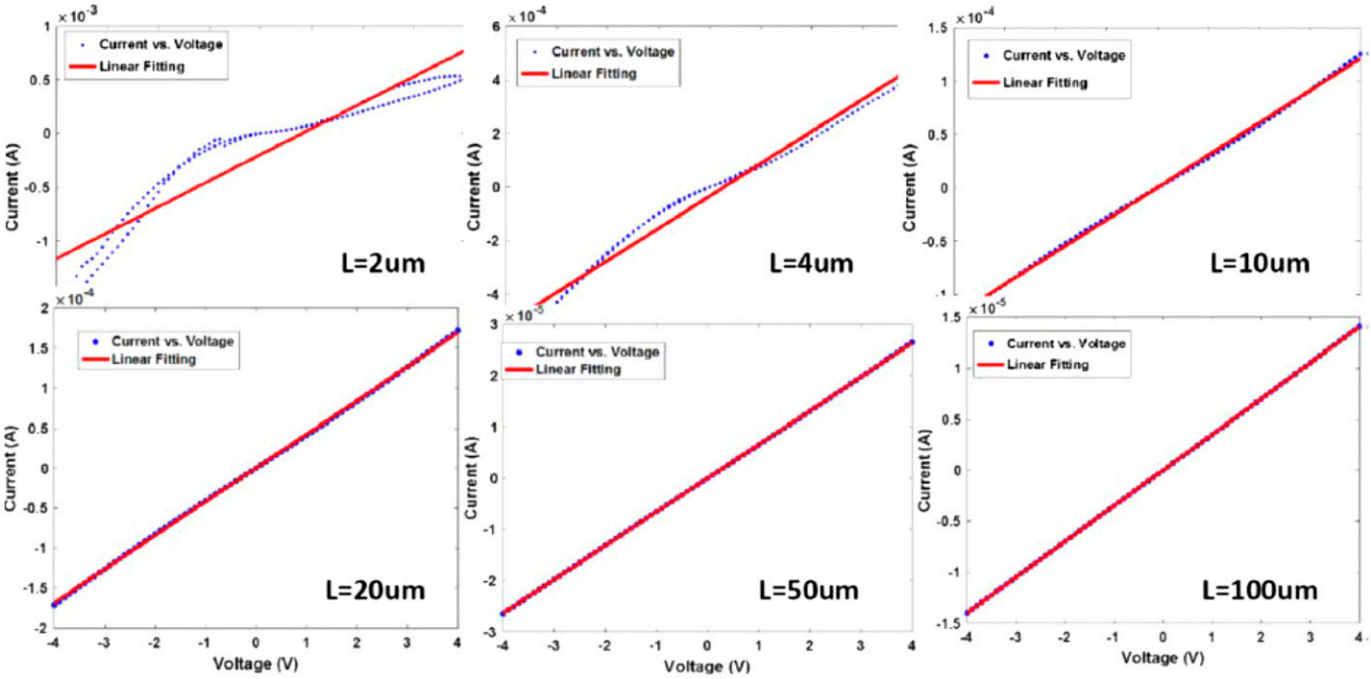

Fig. 4. Measured current-voltage (I-V) curves (blue dot) of CNT resistors against linear fitting model (red line) for resistor lengths from 2 to 100 µm and the width fixed to 40 µm.

resistance attributed by metal-CNT interfaces, L is the resistor length in micrometers (µm), and W_R is the resistor width in micrometers (µm). Although in this practice the critical dimensions, such as resistor length L and width W, are defined using traditional lithography, the materials of CNT films are fully compatible with solution or printing processes that could significantly reduce the manufacturing costs. The fitted linear resistor models and distributions are then incorporated into FHE-PDK for SPICE simulations.

C. Capacitor Models

In addition to CNT resistors, flexible capacitors can also be realized using TFT compatible solution processes, for applications such as filters and equalizers. Here, we use simple top-bottom parallel-plate structures to realize the flexile capacitors with 25-nm thick Al_2O_3 dielectrics as shown in Fig. 3. To further study the process variations of the capacitance, we designed and characterized 431 parallel-plate capacitors, and the measurement results of the unit

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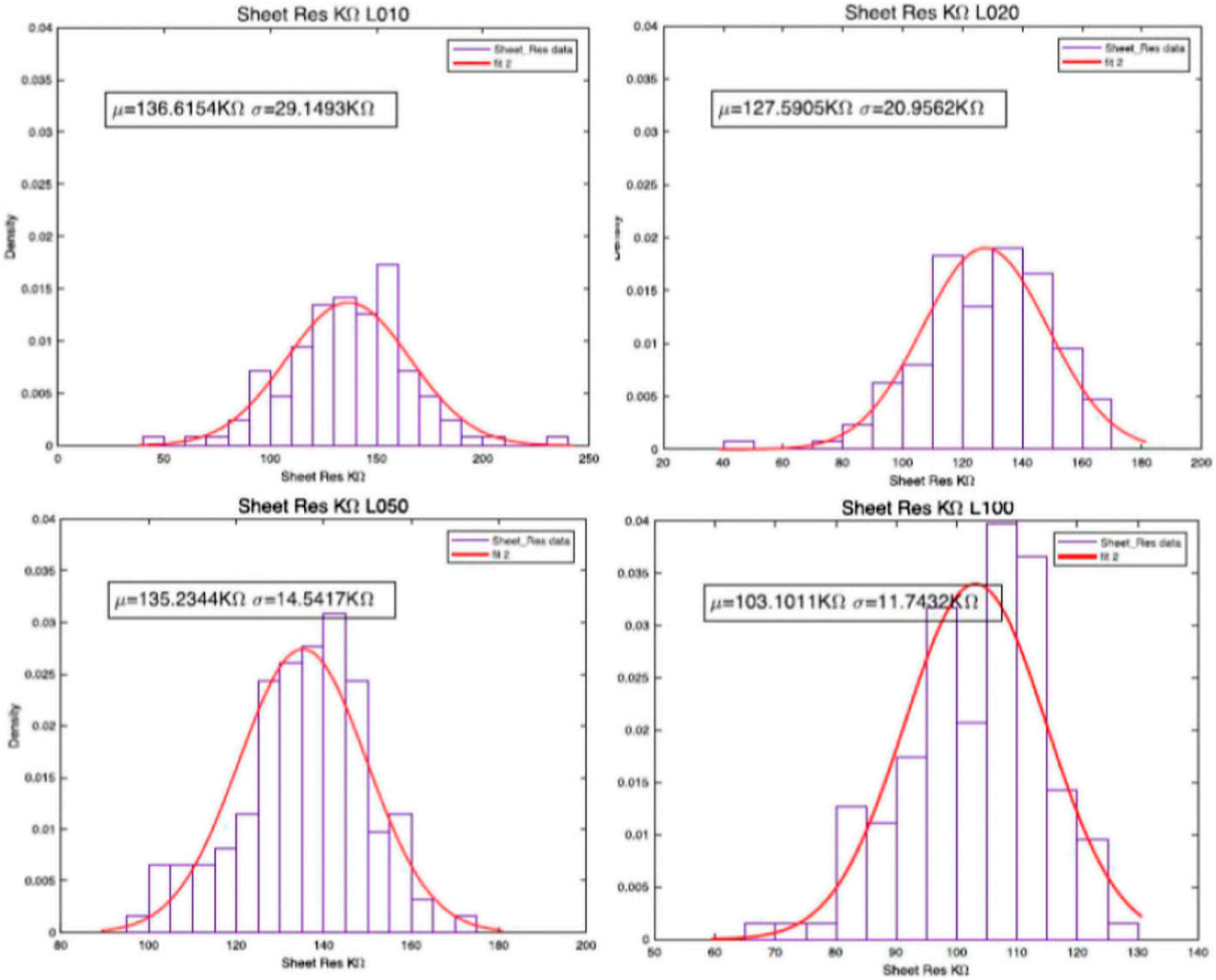

Fig. 5. Measured statistical histograms of CNT resistors for resistor lengths from 10 to 100 μm and the width fixed to 40 μm.

capacitance, i.e., capacitance per unit area, can be found in Fig. 6 for the capacitor width W_C ranging from 45 to 1,415 μm. We can learn that although the standard deviations of the unit capacitance (~3 fF/μm^2) are similar among different capacitor widths W_C, the average unit capacitance increases with larger capacitor widths W_C. We attribute this phenomenon to thinner dielectrics for larger sized capacitors. The 2D spatial analysis in Fig. 7 shows the unit capacitance variations across the same flexible substrate, with which figure we can learn that the unit capacitance is higher at the center of the substrate. This could be attributed to thinner dielectrics at the center of the substrate due to nonuniformity of the dielectrics thickness, which is often seen in solution or printing processes.

Physical Design Verification

To enhance the productivity and reduce the design time required for designing a complex FHE system, we incorporate both silicon CMOS design, i.e., IC-centric, design toolsets and PCB-centric design toolsets for FHE-PDK design, simulation, and physical layout verification flow. We also create experimentally proven design rules of CNT-TFTs, CNT resistors, parallel-plate capacitors using solution processes, and design rules of spiral inductors using screen printing. Based on these design rules, we create technology files using Standard Verification Rule Format, which is tool command language (T-cl) or Python-based formats to perform physical verification using mainstream IC-centric toolsets including Mentor Graphic *Calibre*. The exemplary graphical user interface (GUI) of FHE-PDK using Cadence *Virtuoso* and Mentor Graphic *Calibre* is shown in Fig. 8. The design example is a CNT-TFT variable gain amplifier (VGA). All basic elements such as CNT transistors and parallel-plate capacitors are constructed as P cells. FHE designers, therefore, can simply pull various P cells for elements to build functional circuits that can range from a random pattern generator to a VGA. The simulation models and design rules for given P cells and

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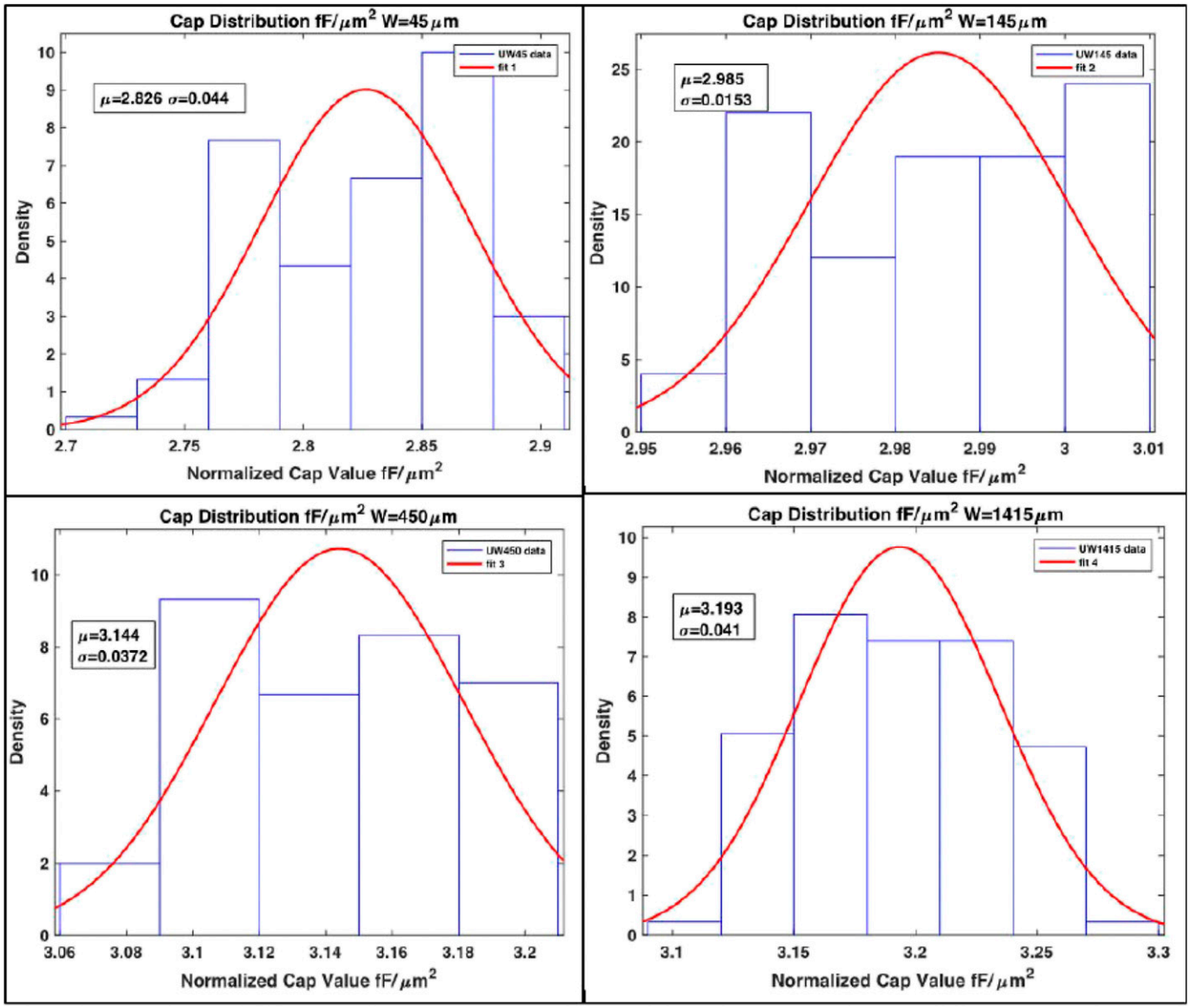

Fig. 6. Measured statistical histograms of unit capacitance for the parallel-plate capacitors. The capacitor widths W_C are from 45 to 445 μm.

design geometries will be loaded automatically in the GUI design environment to perform circuit performance simulations and rule checking for physical layout. In addition to physical design verifications, we also incorporate material properties to create technology files for LPE. Layout-related parasitics in flexible circuits can be significantly larger than silicon chips, which is because of larger physical footprints. Thus, it is essential to include accurate estimates of layout-related parasitics such as resistance and capacitance for postlayout simulations, particularly for high-performance FHE circuits.

In addition to circuit simulations, FHE-PDK is also capable of performing system-level simulations and DRC for a system-level design by incorporating PCB-centric design toolset including Mentor Graphic *HyperLynx* as shown in Fig. 9. The design example here is a heart sound-monitoring patch that consists of microphones, a chip antenna, and Bluetooth system-on-a-chip, all connected by printed traces.

Foundational Design Intellectual Property

A. Digital Logic

Complementary circuits with both p and n-types of transistors are essential in silicon CMOS design to reduce the static power consumption as well as to improve the noise margin. However, certain TFT technologies such as IGZO and amorphous silicon do not offer good complementary devices, which makes low-power and robust TFT digital design difficult. To overcome the design challenges such as the lack of complementary reliable TFTs as well as large process variations for printed and solution-processed TFT circuits, we

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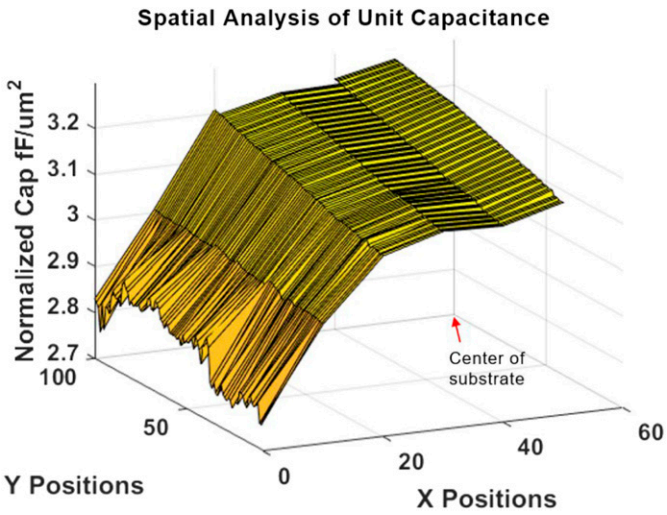

Fig. 7. Measured 2D spatial analysis of the unit capacitance (fF/μm^2) against the substrate dimensions in millimeters (mm).

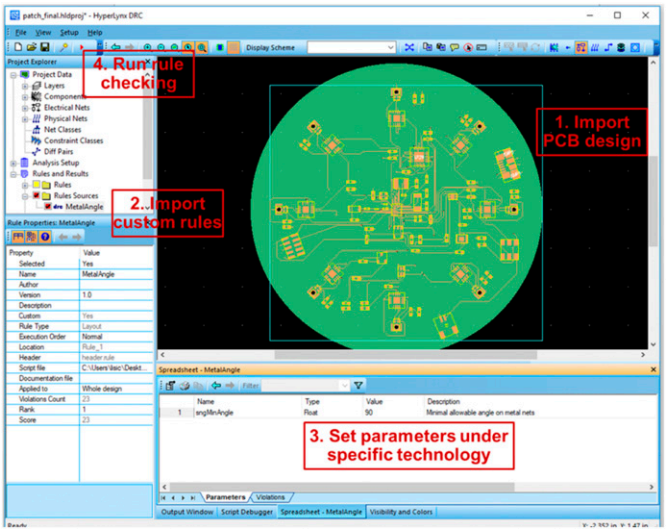

Fig. 9. GUI of FHE-PDK to perform design entry and DRC with *HyperLynx* for a heart-sound monitoring patch.

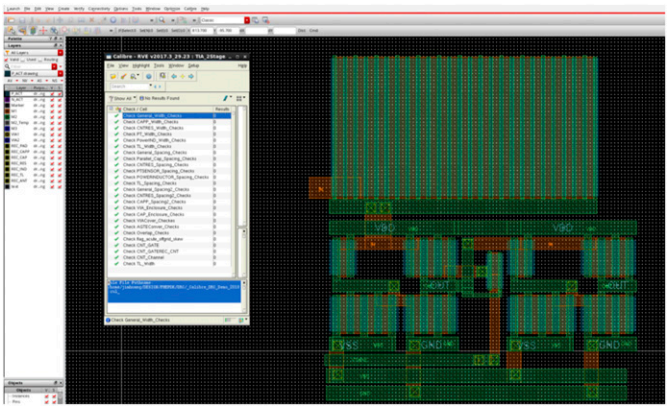

Fig. 8. GUI of FHE-PDK to perform design entry and DRC with *Virtuoso* and *Calibre* for a CNT amplifier.

have previously reported a robust design style *Pseudo-CMOS* [1]. Based on *Pseudo-CMO*S, we have developed a variety of solution process-proven digital logic design IP blocks ranging from a simple inverter, simple logic gates, to linear-feedback shifter registers using CNT-TFTs. With the aid of FHE-PDK, we can now design, simulate, and verify complex TFT digital logic with thousands of CNT-TFTs within a short period of time. The simulated performance and power also provide a good correlation with the fabricated CNT-TFT circuits at low supply voltages (3V) [4].

B. Analog Signal Amplification

In addition to digital logic, we also develop analog design IP blocks such as VGA that is commonly used in sensor interfaces to provide signal amplifications. The schematics and measured input-output waveforms of a 9T-1C VGA are shown in Fig. 10. The voltage gain of the VGA can be tuned by varying the gate voltage VTUNE of the feedback TFT. The physical size of the 9T-1C amplifier is 350 $\times$ 350 μm^2, including a parallel-plate input capacitor to attenuate low-frequency input noises and block the dc current feedback to the input terminal. The gain-bandwidth of the VGA can be tuned from 32 dB gain with 33 KHz bandwidth to 60 dB with 15 KHz bandwidth by varying VTUNE voltages, and the minimum input signal is 5 mV.

Conclusion

In this article, we report our latest development of FHE-PDK for FHE circuit and system design, simulation, and physical design verification. Although FHE is emerging as a viable solution for high-performance low-cost IoT and wearable applications, the device models, design rules, and technology files are still missing to enable the codesign and cosimulations of flexible TFT circuits and thinned silicon chips. FHE-PDK fills these gaps by experimentally validated models, design rules, technology files, P cells, and digital/analog design IP blocks. We believe that FHE-PDK will emerge as the must-have toolbox for FHE designers.

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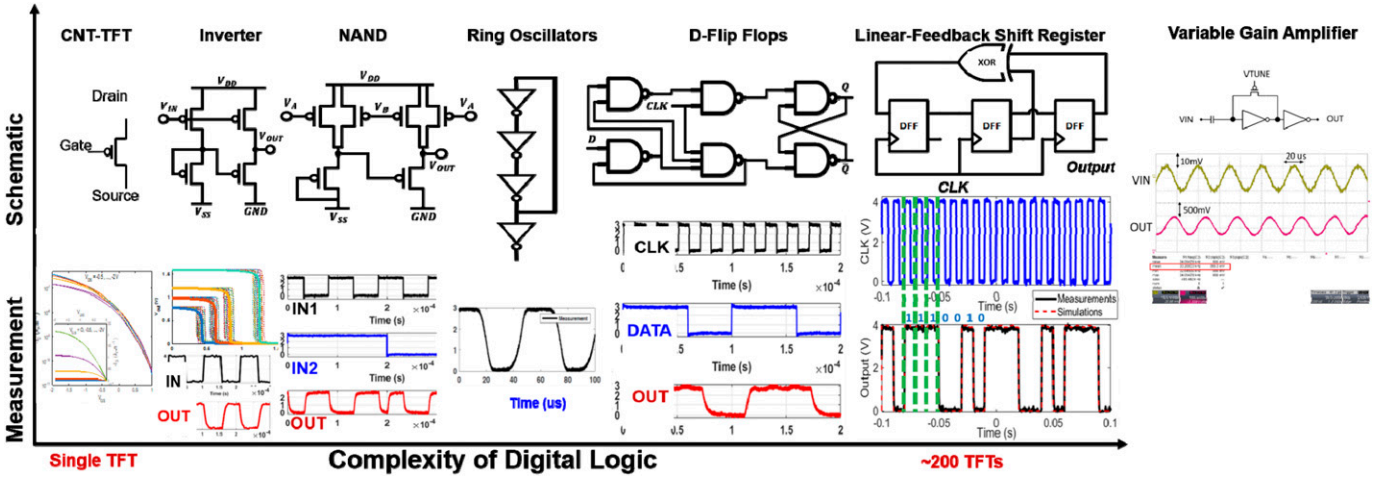

Fig. 10. Overview of foundational design IP blocks using CNT-TFTs for digital logic and analog amplification.

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